MAT1252
Mathematics for Computing

Lecture 01

Bases and Number Representation

Outline

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02. Use of subscripts
03. The octal number system
04. Counting in octal
05. Conversion of octal to decimal
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23. Addition in hexadecimal
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25. Addition of BCD numbers

Lecture's Major Objectives

After completing this section, students should be able to:

- count in each of the decimal, octal, binary and hexadecimal number systems
- convert integers between the number systems
- convert fractions between binary and decimal number systems
- perform addition in each of the number systems
- find the BCD code for a decimal number
- convert a BCD number to its decimal equivalent
- perform BCD additions, making any necessary decimal adjustments

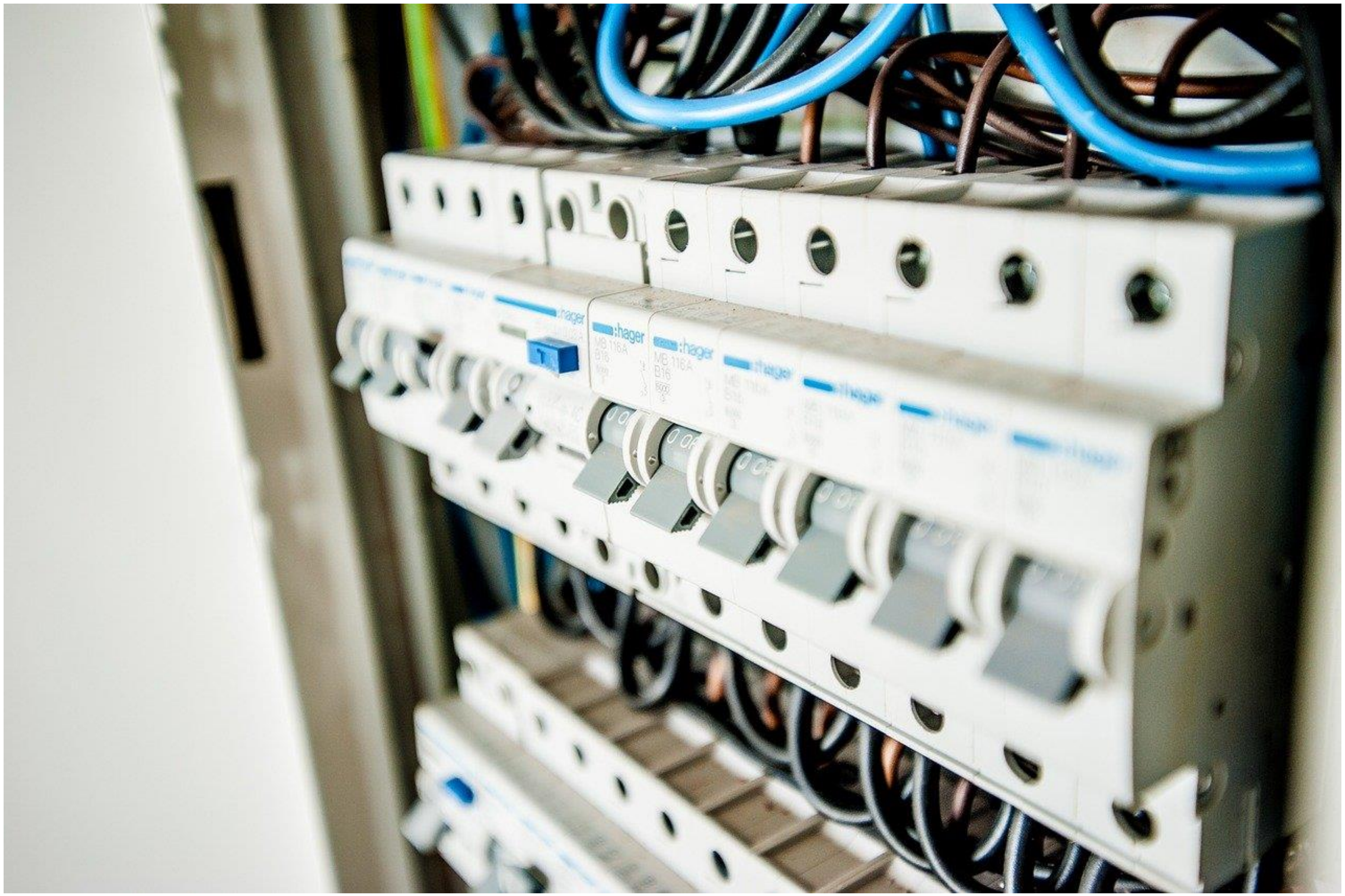
Introduction

Humans count on 10 fingers*…

*in many, but not all, cultures

Introduction

Computers count on switches!



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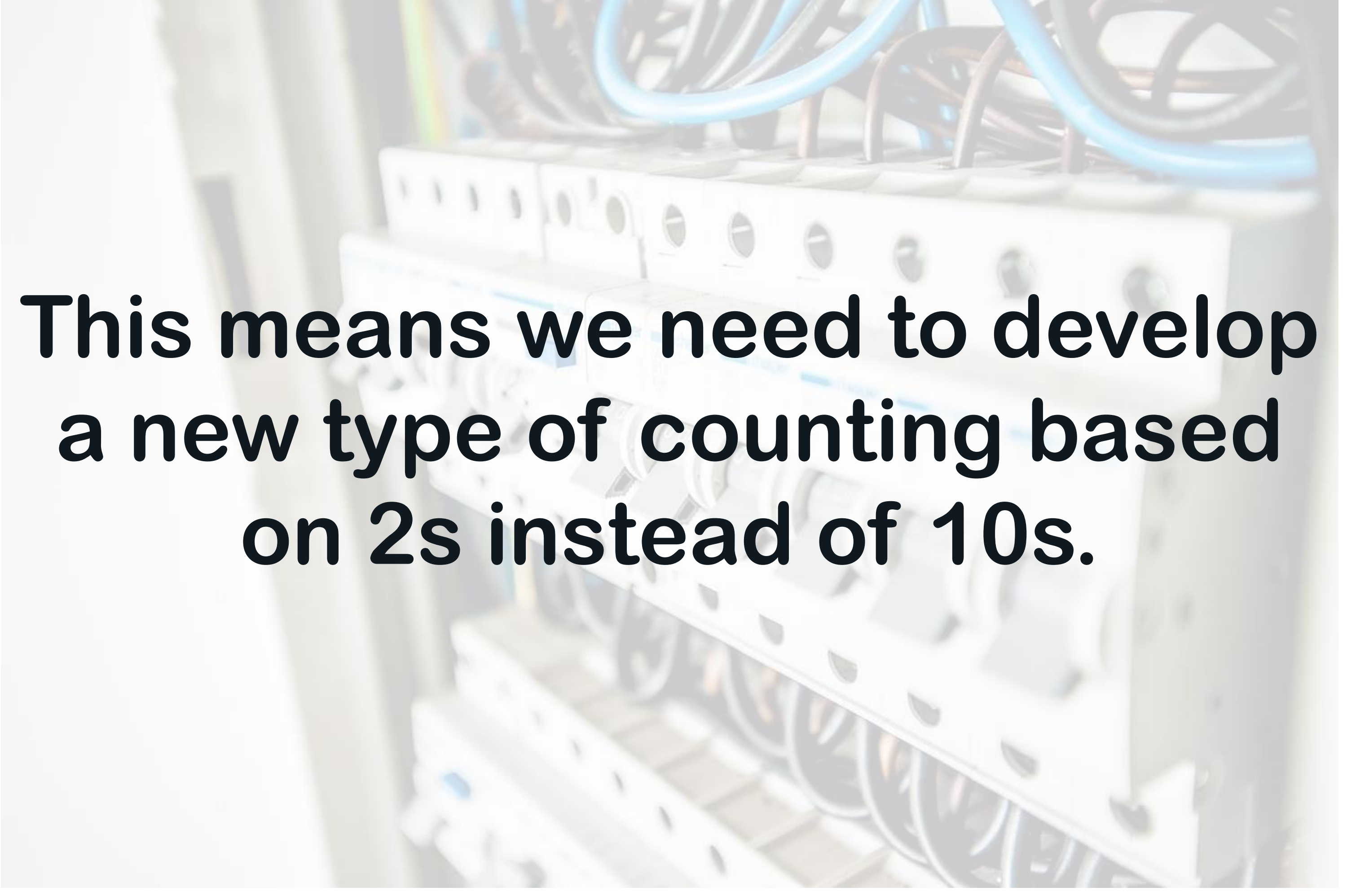


The decimal number system

We are familiar with the decimal number system, which is based on the number **ten**. The decimal system uses **positional notation**.

For example

"354" represents $(3 \times 100) + (5 \times 10) + (4 \times 1)$

$$= (3 \times 10^2) + (5 \times 10) + (4 \times 1)$$

whereas

"2435" represents $(2 \times 10^3) + (4 \times 10^2) + (3 \times 10) + (5 \times 1)$.

In the decimal system we use **ten different symbols** to represent numbers.

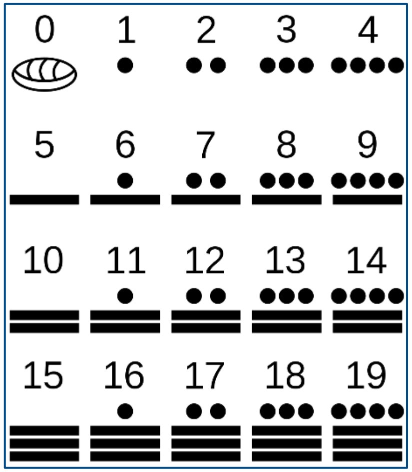
These symbols are: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9

Different number systems

There is nothing special about the decimal number system for representing numbers! Civilisations around the world and throughout history have chosen different bases to count in.

The most common historical bases are those involving 5, 10 and 20.

E.g. the Mayans of South America used base 20 to count. Each column represents successive powers of 20, and they had symbols 0-19 to use in each column.



The most common bases for computers are 2, 8 and 16.

E.g. base 2 is called binary. Each column represents a power of 2, and only the symbols 0 and 1 are needed.

Use of subscripts

In MAT1252, we use different subscripts to indicate which number system we are using, and hence avoid ambiguity.

For example

354_{10}	represents a decimal number (base 10)
354_{8}	represents an octal number (base 8)
1001010_{2}	represents a binary number (base 2)
354_{16}	represents a hexadecimal number (base 16)

The octal number system

The octal system is a **positional system using base 8**.

Octal uses only the 8 symbols 0, 1, 2, 3, 4, 5, 6 and 7.

$(354)_8$ means $(\mathbf{3} \times 8^2) + (\mathbf{5} \times 8) + (\mathbf{4} \times 1)$ which is equal to 236_{10} and

$(435)_8$ means $(\mathbf{4} \times 8^2) + (\mathbf{3} \times 8) + (\mathbf{5} \times 1)$ which is equal to 285_{10}.

Octal number	Meaning	Dec. equivalent
35	3x8 + 5x1	29
54	5x8 + 4x1	44
77	7x8 + 7x1	63
261	$2 \times 8^2 + 6 \times 8 + 1 \times 1$	177

Counting in octal

64	8	1	Dec. Equiv
		0	0
		1	1
		2	2
		...	...
		7	7
	1	0	8
	1	1	9
	...	...	...
	1	7	15
	2	0	16
	2	1	17
	...	...	...
	7	7	63
1	0	0	64
1	0	1	65
...	...	...	...
1	7	7	127
2	0	0	128

Conversion of octal to decimal

To convert from octal to decimal, we recall that each octal digit represents a particular power of 8.

Example 1: Convert 213_8 to decimal.

$$213_8 = (\mathbf{2} \times 8^2) + (\mathbf{1} \times 8) + (\mathbf{3} \times 1)$$

$$= (2 \times 64) + (1 \times 8) + (3 \times 1) = 139_{10}$$

Example 2: Convert 467_8 to decimal.

$$467_8 = (\mathbf{4} \times 8^2) + (\mathbf{6} \times 8) + (\mathbf{7} \times 1)$$

$$= (4 \times 64) + (6 \times 8) + (7 \times 1) = 311_{10}$$

Conversion of octal to decimal

Example 3: Convert 4276_8 to decimal.

Conversion of octal to decimal

Example 3: Convert 4276_8 to decimal.

$$4276_8 = (4 \times 8^3) + (2 \times 8^2) + (7 \times 8) + (6 \times 1)$$

$$= (4 \times 512) + (2 \times 64) + (7 \times 8) + (6 \times 1)$$

$$= 2238_{10}$$

Conversion from decimal to octal

The algorithm is: "Divide by 8 until the answer is 0, then read the remainders in reverse order"

Example: Convert 171_{10} to octal.

When we divide 171 repeatedly by 8, we get:

8	**171**
8	**21 + rem 3**
8	**2 + rem 5**
8	**0 + rem 2**

The remainders, in reverse order, are: 2, 5 and 3.
So $171_{10} = 253_8$

Why does this method work?
$171 = (21 \times 8) + 3$ and
$21 = (2 \times 8) + 5$, so
$171 = [(2 \times 8) + 5] \times 8 + 3$
therefore $171 = (2 \times 8^2) + (5 \times 8) + 3$

Conversion from decimal to octal

Example: Convert 278_{10} to octal.

8	278
8	34 + rem 6
8	4 + rem 2
8	0 + rem 4

So $278_{10} = 426_8$

[Check: $426_8 = (4 \times 64) + (2 \times 8) + (6 \times 1) = 278_{10}$]

The binary number system

The binary system is a **positional system using base 2**.
In this system there are only two symbols, 0 and 1.

So, for example, the binary number 101 represents

$(\mathbf{1} \times 2^2) + (\mathbf{0} \times 2) + (\mathbf{1} \times 1) = 5$ in decimal

and the binary number 1101 represents

$(\mathbf{1} \times 2^3) + (\mathbf{1} \times 2^2) + (\mathbf{0} \times 2) + (\mathbf{1} \times 1) = 13$ in decimal

Here are some binary numbers and their decimal equivalents:

Binary number	Meaning	Dec. equivalent
111	4+2+1	7
1001	8+0+0+1	9
110011	32+16+0+0+2+1	51
10000000	128+0+...+0	128

Counting in binary

16	8	4	2	1	Dec. equiv
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	2
0	0	0	1	1	3
0	0	1	0	0	4
0	0	1	0	1	5
0	0	1	1	0	6
0	0	1	1	1	7
0	1	0	0	0	8
0	1	0	0	1	9
0	1	0	1	0	10
0	1	0	1	1	11
0	1	1	0	0	12
0	1	1	0	1	13
0	1	1	1	0	14
0	1	1	1	1	15
1	0	0	0	0	16
1	0	0	0	1	17
1	0	0	1	0	18
1	0	0	1	1	19
1	0	1	0	0	20
1	0	1	0	1	21
1	0	1	1	0	22
1	0	1	1	1	23
1	1	0	0	0	24
1	1	0	0	1	25
1	1	0	1	0	26
1	1	0	1	1	27
1	1	1	0	0	28
1	1	1	0	1	29
1	1	1	1	0	30
1	1	1	1	1	31

Conversion from binary to decimal

Method 1: Write each number as the sum of powers of 2

Example 1: Convert 10111001_2 to decimal.

$$10111001_2 = 2^7 + 2^5 + 2^4 + 2^3 + 1$$
$$= 128 + 32 + 16 + 8 + 1 = 185_{10}$$

Example 2: Convert 101111011_2 to decimal.

Example 3: Convert 111100_2 to decimal.

Conversion from binary to decimal

Method 1: Write each number as the sum of powers of 2

Example 1: Convert 10111001_2 to decimal.

$$10111001_2 = 2^7 + 2^5 + 2^4 + 2^3 + 1$$
$$= 128 + 32 + 16 + 8 + 1 = 185_{10}$$

Example 2: Convert 101111011_2 to decimal.

$$101111011_2 = 2^8 + 2^6 + 2^5 + 2^4 + 2^3 + 2 + 1$$
$$= 256 + 64 + 32 + 16 + 8 + 2 + 1 = 379_{10}$$

Example 3: Convert 111100_2 to decimal.

$$111100_2 = 2^5 + 2^4 + 2^3 + 2^2$$
$$= 32 + 16 + 8 + 4 = 60_{10}$$

Conversion from binary to decimal

Method 2

1. Convert to octal by separating groups of three binary digits, starting at the units digit. Then replace each group of three binary digits by its octal equivalent.

0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

2. Convert from octal to decimal.

Conversion from binary to decimal

Example 1 (again): Convert 10111001_2 to decimal.

Example 2 (again): Convert 101111011_2 to decimal.

Example 3 (again): Convert 111100_2 to decimal.

Conversion from binary to decimal

Example 1 (again): Convert 101111001_2 to decimal.

$10 \mid 111 \mid 001 = 271_8 = (2 \times 64) + (7 \times 8) + 1 = 185_{10}$

2 7 1

Example 2 (again): Convert 101111011_2 to decimal.

$101\ 111\ 011 = 573_8 = (5 \times 64) + (7 \times 8) + 3 = 379_{10}$

Example 3 (again): Convert 111100_2 to decimal.

$111\ 100 = 74_8 = (7 \times 8) + 4 = 60_{10}$

Conversion from decimal to binary

Method 1: Divide by 2 until the answer is 0, then read the remainders in reverse order.

Example 1: Convert 58_{10} to binary.

2	**58**
2	**29 + rem 0**
2	**14 + rem 1**
2	**7 + rem 0**
2	**3 + rem 1**
2	**1 + rem 1**
	0 + rem 1

The remainders, in reverse order, are 1, 1, 1, 0, 1 and 0.

Therefore $58_{10} = 111010_2$

[Check: $111010_2 = 32 + 16 + 8 + 0 + 2 + 0 = 58_{10}$]

Conversion from decimal to binary

An alternative (and much quicker) method is to **use octal**.

First you need to again remember the binary equivalents of the numbers 0, 1, 2, …, 7.

These are:

0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Conversion from decimal to binary

Method 2: First convert the **decimal number to octal**, and then replace each octal digit by its binary equivalent.

Example 1 (again): Convert 58_{10} to binary

First convert to octal:

8	58
8	7 + rem 2
8	0 + rem 7

Reading the remainders in reverse order, we find $58_{10} = 72_8$

Now replace each octal digit with the corresponding binary equivalent.

(In this case replace 7 by 111 and replace 2 by 010)

So $58_{10} = 72_8 = 111010_2$

Conversion from decimal to binary

Example 2: Convert 370_{10} to binary.

First convert to octal:

8	**370**
8	**46 + rem 2**
8	**5 + rem 6**
8	**0 + rem 5**

Now replace each octal digit with the corresponding binary equivalent.

So $370_{10} = 562_8 = 101110010_2$

The hexadecimal number system

This system is based on the number 16. Its main purpose is to help us to cope with **long strings of binary digits**. For example, suppose we had to remember the binary number:

00101101111100111000110001001011

We would find it much easier to work with its hexadecimal equivalent

2DF38C4B

16^3	16^2	16	1	Dec. equiv.
		1	5	21
	1	0	3	259
		5	7	87
1	0	0	0	4096
		4	6	70

The hexadecimal number system

The hexadecimal system is a **positional system using base 16.** Hexadecimal uses 16 symbols. The symbols 0, 1, 2, …, 9 have the same meaning as in decimal, and we need six extra symbols to represent these six numbers. We use A, B, C, D, E and F:

A represents the decimal number 10,
B represents the decimal number 11,
C represents the decimal number 12,
D represents the decimal number 13,
E represents the decimal number 14, and
F represents the decimal number 15.

"5C" in hex represents $(5 \times 16) + 12 = 92$ in decimal.
"1AF" in hex represents $(1 \times 16^2) + (10 \times 16) + (15 \times 1) = 431$ in decimal.

The hexadecimal number system

Here are some more hexadecimal numbers and their decimal equivalents:

(Note: $16^2 = 256$ and $16^3 = 4096$)

Hex. number	Meaning	Dec. equiv.
2B7	2x256 + 11x16 + 7x1	695
496	4x256 + 9x16 + 6x1	1174
FE	15x16 + 14x1	254
CB4D	12x4096 + 11x256 + 4x16 +13x1	52045

Counting in hexadecimal

16^3	16^2	16	1	Dec. equiv.
0	0	0	0	0
0	0	0	1	1
0	0	0	2	2
...	...	...	...	...
0	0	0	9	9
0	0	0	A	10
...	...	...	...	...
0	0	0	F	15
0	0	1	0	16
0	0	1	1	17
...	...	...	...	...
0	0	1	D	29
0	0	1	E	30
0	0	1	F	31
0	0	2	0	32
...	...	...	...	...
0	0	9	F	159
0	0	A	0	160
...	...	...	...	...
0	0	F	F	255
0	1	0	0	256
...	...	...	...	...
F	F	F	F	65535

Conversion of hexadecimal numbers to binary

First we need to be familiar with the binary equivalents of the hexadecimal numbers 0, …, F. These are:

0	**0000**
1	**0001**
2	**0010**
3	**0011**
4	**0100**
5	**0101**
6	**0110**
7	**0111**

8	**1000**
9	**1001**
A	**1010**
B	**1011**
C	**1100**
D	**1101**
E	**1110**
F	**1111**

Then it is simply a matter of replacing each hexadecimal digit by the corresponding four binary digits.

Conversion of binary numbers to hexadecimal

To convert a binary integer to hexadecimal, **separate binary digits into groups of four**, starting at the units digit, and then replace each group by its hexadecimal equivalent.

Example 1:

Convert $10111110101011001 1_2$ to hexadecimal.

$$10\ \ 1111\ \ 1010\ \ 1011\ \ 0011_2 = 2FAB3_{16}$$

Example 2:

Convert 1111100001011101101_2 to hexadecimal.

$$111\ \ 1100\ \ 0010\ \ 1110\ \ 1101_2 = 7C2ED_{16}$$

Conversion between hexadecimal and decimal

Sometimes it is convenient to be able to convert from simple hexadecimal numbers to decimal, or vice versa.

For conversions such as these it helps to know, or at least be familiar with, the multiples of 16.

Examples:

1. The hex number 3C means $(3 \times 16) + 12 = 60$ in decimal.

2. The hex number 54 means $(5 \times 16) + 4 = 84$ in decimal.

3. The hex number F6 means $(15 \times 16) + 6 = 246$ in decimal.

Conversion between hexadecimal and decimal

4. To convert the decimal number 43 to hexadecimal:

    43 divided by 16?

    $2 \times 16 = 32$, so

    43 divided by 16 gives 2 with remainder 11

    So $43 = (2 \times 16) + 11$

    Therefore 43 in decimal becomes 2B in hex.

5. To convert the decimal number 77 to hexadecimal:

    77 divided by 16?

    $4 \times 16 = 64$, so

    77 divided by 16 gives 4 with remainder 13

    So $77 = (4 \times 16) + 13$

    Therefore 77 in decimal becomes 4D in hex.

Conversion between hexadecimal and decimal

6. To convert the decimal number 103 to hexadecimal:

   103 divided by 16?

   $6 \times 16 = 96$, so

   103 divided by 16 gives 6 with 7 remainder

   So $103 = (6 \times 16) + 7$

   Therefore 103 in decimal becomes 67 in hex.

Representation of binary fractions

Now we consider fractions in binary. The concepts are analogous to those in the decimal system.

In the decimal system, the columns to the right of the decimal point represent

$$\frac{1}{10}, \frac{1}{100}, \frac{1}{1000}, \ldots \text{ and so on.}$$

Here are some decimal numbers with both integer parts and fractional parts:

...	100	10	1		$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$	...	Meaning
		4	3	.	2	5			43.25
			6	.	0	7	1		6.071
	9	8	5	.	3	0	2		985.302
			0	.	9	9	0		0.99
			0	.	3	2	4		0.324

Representation of binary fractions

In decimal, for example

$$0.324 \ \text{means} \ \left(3 \times \frac{1}{10}\right) + \left(2 \times \frac{1}{10^2}\right) + \left(4 \times \frac{1}{10^3}\right) = \frac{324}{1000}$$

Notice that the fraction 0.324 has 3 places after the decimal point so the last place represents $\frac{1}{10^3}$.

In the binary system, the columns to the right of the point represent $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$ … and so on. In other words, $\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \frac{1}{2^4}, \ldots$

Representation of binary fractions

Here are some binary numbers with both integer parts and fractional parts:

...	4	2	1		$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	...	Binary meaning	Decimal meaning
			1	.	1					1.1	1.5
			0	.	1	1				0.11	0.75
		1	1	.	0	1				11.01	3.25
	1	0	1	.	1	1	1			101.111	5.875
			0	.	1	1	0	1		0.1101	0.8125

Representation of binary fractions

In the binary system, for example,

0.1101_2 means $\frac{1}{2} + \frac{1}{4} + \frac{0}{8} + \frac{1}{16}$ or $\frac{13}{16}$ = 0.8125 in decimal.

Here the fraction 0.1101 has 4 places after the decimal point so the last place represents $\frac{1}{2^4}$.

Example

0.10011_2 means $\frac{1}{2} + \frac{1}{16} + \frac{1}{32}$ or $\frac{19}{32}$ or 0.59375 in decimal

Converting binary fractions to decimal

A simple method is:

1. Convert the fractional part (i.e. the number after the decimal place) to a decimal number.
2. Divide your answer by the appropriate power of 2 as given by the number of places in the fractional part.

Example 1: Convert 0.110011_2 to decimal.

1. The number after the point is $(110011)_2$ which is 51 in decimal.
2. The fraction has 6 places, so we must divide by $2^6 = 64$.

Therefore $0.110011_2 = \dfrac{51}{64} = 0.796875_{10}$

Converting binary fractions to decimal

Example 2: Convert 0.11101_2 to decimal.

1. The number after the point is $(11101)_2 = 29_{10}$
2. The fraction has 5 places, so we must divide by $2^5 = 32$.

Therefore $0.11101_2 = \dfrac{29}{32} = 0.90625_{10}$

Example 3: Convert 0.001010101_2 to decimal.

1. The number after the point is $(1010101)_2 = 85_{10}$.
2. The fraction has 9 places, so we must divide by $2^9 = 512$.

Therefore $0.001010101_2 = \dfrac{85}{512} = 0.166015625_{10}$

Converting binary fractions to decimal

Example 4: Convert 0.00000101_2 to decimal.

1. The number 101 in binary is 5 in decimal.
2. The fraction has 8 places, so we must divide by $2^8 = 256$.

Therefore $0.00000101_2 = \frac{5}{256} = 0.01953125_{10}$

Converting decimal fractions to binary

Method

Repeatedly multiply the digits to the **right of the decimal point** by 2, until only zeros remain. Then read the integer parts of the successive answers, from top to bottom.

Example 1: Convert the decimal fraction 0.34375 to binary.

$$
\begin{array}{r}
0.34375 \\
\underline{\times \quad 2} \\
0.68750 \\
\underline{\times \quad 2} \\
1.37500 \\
\underline{\times \quad 2} \\
0.75000 \\
\underline{\times \quad 2} \\
1.50000 \\
\underline{\times \quad 2} \\
1.00000
\end{array}
$$

So $0.34375_{10} = 0.01011_2$

Converting decimal fractions to binary

Example 2: Convert the decimal fraction 0.453125 to binary.

0.453125
2
0.906250
2
1.812500
2
1.625000
2
1.250000
2
0.500000
2
1.000000

So $0.453125_{10} = 0.011101_2$

Converting decimal fractions to binary

Example 3: Convert the decimal fraction 0.65625 to binary.

$$
\begin{array}{r}
0.65625 \\
\underline{2} \\
\mathbf{1}.31250 \\
\underline{2} \\
\mathbf{0}.62500 \\
\underline{2} \\
\mathbf{1}.25000 \\
\underline{2} \\
\mathbf{0}.50000 \\
\underline{2} \\
\mathbf{1}.00000
\end{array}
$$

So $0.65625_{10} = 0.10101_2$

Summary of number conversion

Conversion	*Method*	*Example*
Binary to		
Octal	Substitution	$10111011001_2 = 10\ 111\ 011\ 001_2 = 2731_8$
Hexadecimal	Substitution	$10111011001_2 = 101\ 1101\ 1001_2 = 5D9_{16}$
Decimal	Summation	$10111011001_2 = 1x1024 + 0x512 + 1x256 + 1x128 + 1x64 + 0x32 + 1x16 + 1x8 + 0x4 + 0x2 + 1x1 = 1497_{10}$
Octal to		
Binary	Substitution	$1234_8 = 001\ 010\ 011\ 100_2$
Hexadecimal	Substitution	$1234_8 = 001\ 010\ 011\ 100_2 = 0010\ 1001\ 1100_2 = 29C_{16}$
Decimal	Summation	$1234_8 = 1x512 + 2x64 + 3x8 + 4x1 = 668_{10}$

Summary of number conversion

Conversion	*Method*	*Example*
Hexadecimal to		
Binary	Substitution	$C0DE_{16} = 1100\ 0000\ 1101\ 1110_2$
Octal	Substitution	$C0DE_{16} = 1100\ 0000\ 1101\ 1110_2$ $= 1\ 100\ 000\ 011\ 011\ 110_2 = 140336_8$
Decimal	Summation	$C0DE_{16} = 12x4096 + 0x256 + 13x16 + 14x1 = 49374_{10}$

Summary of number conversion

Conversion	Method	Example
Decimal to		
Binary	Division	$108_{10} \div 2 = 54$ reminder 0 (LSB) “least significant bit” $\div 2 = 27$ reminder 0 $\div 2 = 13$ reminder 1 $\div 2 = 6$ reminder 1 $\div 2 = 3$ reminder 0 $\div 2 = 1$ reminder 1 $\div 2 = 0$ reminder 1 (MSB) “most significant bit” $108_{10} = 1101100_2$
Octal	Division	$108_{10} \div 8 = 13$ reminder 4 (LSB) $\div 8 = 1$ reminder 5 $\div 8 = 0$ reminder 1 (MSB) $108_{10} = 154_8$
Hexadecimal	Division	$108_{10} \div 16 = 6$ reminder 12 (LSB) $\div 16 = 0$ reminder 6 (MSB) $108_{10} = 6C_{16}$

Arithmetic using other number systems

We are mainly concerned with the addition of two numbers. An example in the decimal system:

$$\begin{array}{cccc} \mathbf{1} & \mathbf{1} & & \\ 2 & 8 & 5 & \\ 4 & 3 & 8 & + \\ \hline 7 & 2 & 3 & \end{array}$$

In 1st column, 8 + 5 = 13, so we "put down 3 and carry the 1".
In 2nd column, 3 + 8 + 1 = 12, we "put down 2 and carry the 1".
In 3^{rd} column, 1 + 2 + 4 = 7.

Addition in octal

Remember that in octal the digits represent 1, 8, 8^2, 8^3, and so on.

So, if the sum of the two digits in the 1's column is 8 or more, there will be a carry of 1 into the next column, and so on.

Example 1 :

$$
\begin{array}{cccc}
1 & 1 & & \\
2 & 7 & 5 & \\
3 & 6 & 7 & + \\
\hline
6 & 6 & 4 &
\end{array}
$$

In 1st column, $7_8 + 5_8$ (= 12_{10} = 8 + 4) = 14_8 , so the carry is 1.

In 2nd column, $6_8 + 7_8 + 1_8 = 16_8$, so the carry is 1.

In 3rd column, $1_8 + 2_8 + 3_8 = 6_8$.

Addition in octal

Example 2:

$$
\begin{array}{ccccc}
\color{blue}{1} & \color{blue}{1} & \color{blue}{1} & & \\
4 & 5 & 3 & 6 & \\
2 & 2 & 4 & 5 & + \\
\hline
7 & 0 & 0 & 3 &
\end{array}
$$

In 1st column, $6_8 + 5_8 = 13_8$, so the carry is 1.

In 2nd column, $4_8 + 3_8 + 1_8 = 10_8$, so the carry is 1.

In 3rd column, $2_8 + 5_8 + 1_8 = 8_8 = 10_8$, so the carry is 1.

In 4th column, $2_8 + 4_8 + 1_8 = 7_8$.

Addition in binary

Example 1:

1	1	1	1		
	1	0	1	1	
		1	0	1	+
1	0	0	0	0	

In 1st column, 1 + 1 = 10, so the carry is 1.

In 2nd column, 0 +1 + 1 = 10, so the carry is 1.

In 3rd column, 1 + 0 + 1 = 10 , so the carry is 1.

In 4th column, 1 + 1 = 10 , so the carry is 1.

Addition in binary

Example 2:

$$
\begin{array}{cccccccc}
\color{blue}{1} & \color{blue}{1} & \color{blue}{1} & \color{blue}{1} & \color{blue}{1} & & & \\
 & 1 & 1 & 1 & 0 & 1 & & \\
 & 1 & 0 & 1 & 1 & 1 & + & \\
\hline
1 & 1 & 0 & 1 & 0 & 0 & &
\end{array}
$$

In 1st column, 1 + 1 = 10, so the carry is 1.

In 2nd column, 1 + 0 + 1 = 10, so the carry is 1.

In 3rd column, 1 + 1 + 1 = 11 , so the carry is 1.

In 4th column, 0 + 1 + 1 = 10, so the carry is 1.

In 5th column, 1 + 1 + 1 = 11 , so the carry is 1.

Addition in binary

Example 3:

1	0	0	1	1	0	1	0	1	
		1	0	1	1	1	1	0	+

Addition in binary

Example 3:

	1	1	1	1	1				
1	0	0	1	1	0	1	0	1	
		1	0	1	1	1	1	0	+
1	1	0	0	1	0	0	1	1	

Addition in hexadecimal

In hexadecimal the digits represent 1, 16, 16^2, 16^3, and so on.

So, if the sum of the two digits in the 1's column is **16 or more**, there will be a carry of 1 into the next column, and so on.

Examples:

6	1	
3	7	+
9	8	

3	A	
C	5	+
F	F	

	8	5	
	9	D	+

	B	9	
	7	E	+

	A	B	
	C	D	+

	C	C	
	4	D	+

Addition in hexadecimal

In hexadecimal the digits represent 1, 16, 16^2, 16^3, and so on.

So, if the sum of the two digits in the 1's column is **16 or more**, there will be a carry of 1 into the next column, and so on.

Examples:

$$\begin{array}{cccc} & 6 & 1 & \\ & 3 & 7 & + \\ \hline & 9 & 8 & \end{array} \qquad \begin{array}{cccc} & 3 & A & \\ & C & 5 & + \\ \hline & F & F & \end{array} \qquad \begin{array}{cccc} 1 & 1 & & \\ & 8 & 5 & \\ & 9 & D & + \\ \hline 1 & 2 & 2 & \end{array}$$

$$\begin{array}{cccc} 1 & 1 & & \\ & B & 9 & \\ & 7 & E & + \\ \hline 1 & 3 & 7 & \end{array} \qquad \begin{array}{cccc} 1 & 1 & & \\ & A & B & \\ & C & D & + \\ \hline 1 & 7 & 8 & \end{array} \qquad \begin{array}{cccc} 1 & 1 & & \\ & C & C & \\ & 4 & D & + \\ \hline 1 & 1 & 9 & \end{array}$$

Binary Coded Decimal code (BCD)

In BCD code, each digit of a decimal number is represented by a 4-bit binary "code group". This is easier than decimal for computers.

Since there are 10 decimal digits, the 4-bit binary numbers for 0, 1, …, 9

are used for BCD representation, as shown in the table.

decimal digit	BCD code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Binary Coded Decimal code (BCD)

The remaining 4-bit binary numbers 1010, 1011, 1100, 1101, 1110 and 1111, which represent A, B C, D, E and F in hexadecimal, do not correspond to any decimal digit, and are called **"invalid BCD code groups"**.

To convert from decimal to BCD we replace each decimal digit by its 4-bit binary code group.

Example 1:

For the decimal number 37, its BCD representation is 0011 0111

$$\begin{array}{cc} 0011 & 0111 \\ \downarrow & \downarrow \\ 3 & 7 \end{array}$$

Note this is different to the binary representation, which is 100101.

Binary Coded Decimal code (BCD)

Example 2:

For the decimal number 65, the BCD representation is

$$\begin{array}{cc} 0110 & 0101 \\ \downarrow & \downarrow \\ 6 & 5 \end{array}$$

And the binary representation is 1000001.

Example 3:

For the decimal number 123, the BCD representation is

$$\begin{array}{ccc} 0001 & 0010 & 0011 \\ \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 \end{array}$$

And the binary representation is 1111011.

Binary Coded Decimal code (BCD)

Example 4:

Fo the decimal number 478, the BCD representation is:

4	7	8
↓	↓	↓
0100	0111	1000

Binary Coded Decimal code (BCD)

Example 5:

Consider the decimal number 345602. Its BCD representation is as shown:

3	4	5	6	0	2
↓	↓	↓	↓	↓	↓
0011	0100	0101	0110	0000	0010

Now we want to convert from BCD code back to decimal.
This is done by

1. Separating the BCD number into groups of 4 bits
2. Replacing each group by the corresponding decimal digit.

Binary Coded Decimal code (BCD)

Example 6:

Convert the BCD number 0101011100000001 into its decimal equivalent

0101	0111	0000	0001
↓	↓	↓	↓
5	7	0	1

Example 7:

The binary sequence 0100 1011 0011 is NOT a valid BCD representation, because when we try to convert it into its decimal equivalent, we find

0100	1011	0011
↓	↓	↓
4	?	3

We say 1011 is an **invalid BCD code group**.

Addition of BCD numbers

In the following discussion about addition of BCD numbers it is much easier to use hexadecimal rather than binary representation.

So valid BCD code groups are represented by the hex numbers 0, 1, 2, …, 9, while the hex numbers A, B, C, D, E and F represent invalid BCD code groups.

When a microprocessor adds BCD numbers, it is really adding binary (or hex) numbers, but the final output must look like the result of decimal addition.

Binary (or hex) additions may or may not look like decimal additions.

Addition of BCD numbers

These three hex additions can be interpreted as correct decimal additions:

$$\begin{array}{rrl} 3 & 5 & \\ 2 & 4 & + \\ \hline 5 & 9 & \end{array} \qquad \begin{array}{rrl} 1 & 6 & \\ 3 & 1 & + \\ \hline 4 & 7 & \end{array} \qquad \begin{array}{rrl} 4 & 3 & \\ 5 & 2 & + \\ \hline 9 & 5 & \end{array}$$

However the following three hex additions do not give correct decimal answers:

$$\begin{array}{rrl} 2 & 4 & \\ 3 & 6 & + \\ \hline 5 & A & \end{array} \qquad \begin{array}{rrl} 7 & 5 & \\ 8 & 7 & + \\ \hline F & C & \end{array} \qquad \begin{array}{rrl} 3 & 8 & \\ 3 & 9 & + \\ \hline 7 & 1 & \end{array}$$

Addition of BCD numbers

When two BCD code groups are added, there are three possible types of outcomes as illustrated by the following three cases. Remember the additions are all hex:

(i)	(ii)	(iii)
3	6	9
4 +	7 +	8 +
7	D	1 1

In case (i), if the result is interpreted as decimal, a correct answer is obtained.

In case (ii) an invalid BCD code group is produced so the result cannot be interpreted as decimal.

In case (iii), the result, if interpreted as decimal, gives a wrong answer.

Addition of BCD numbers

BCD addition means doing addition in hex but adjusting the answer so that it makes sense when the inputs are interpreted in decimal.

The hexadecimal addition of the inputs and the decimal adjustments occur inside the machine and are therefore hidden from the user.

Although all the calculations are in hex (or binary), as far as the user can see, the operation is simply decimal addition.

The steps involved in BCD addition are shown in the following slide. Notice that the first addition and then the decimal adjustment on each column are completed **before** moving to the next column.

Addition of BCD numbers

The procedure for BCD addition

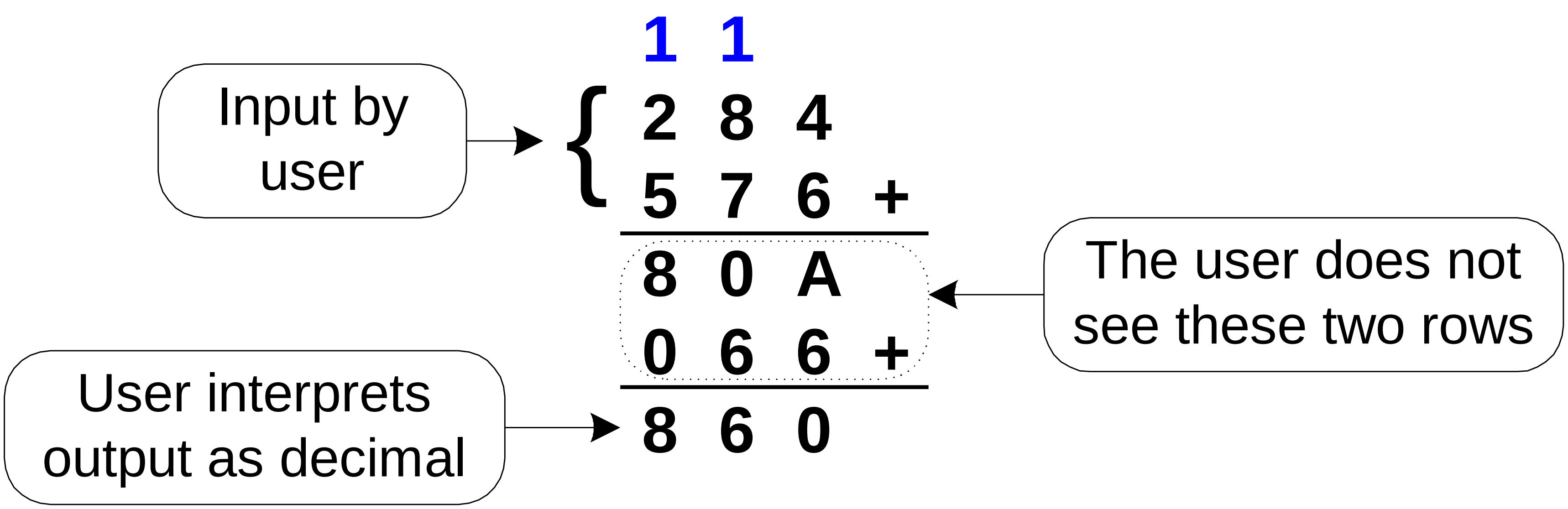
Start at the right hand column (note: we work on one column at a time):

- Add the two digits and transfer any carry to the next column;
- Make the decimal adjustment by adding 6 or 0, according to the rule:
  - If the sum of the two digits is **more than 9**, add 6
  - Otherwise add 0.
- Add the two newly obtained digits of the column and transfer any carry to the next column.
- Move to the next column and repeat the above operations, until all columns have been processed.

Note that a carry may come from the first addition or from the second addition.

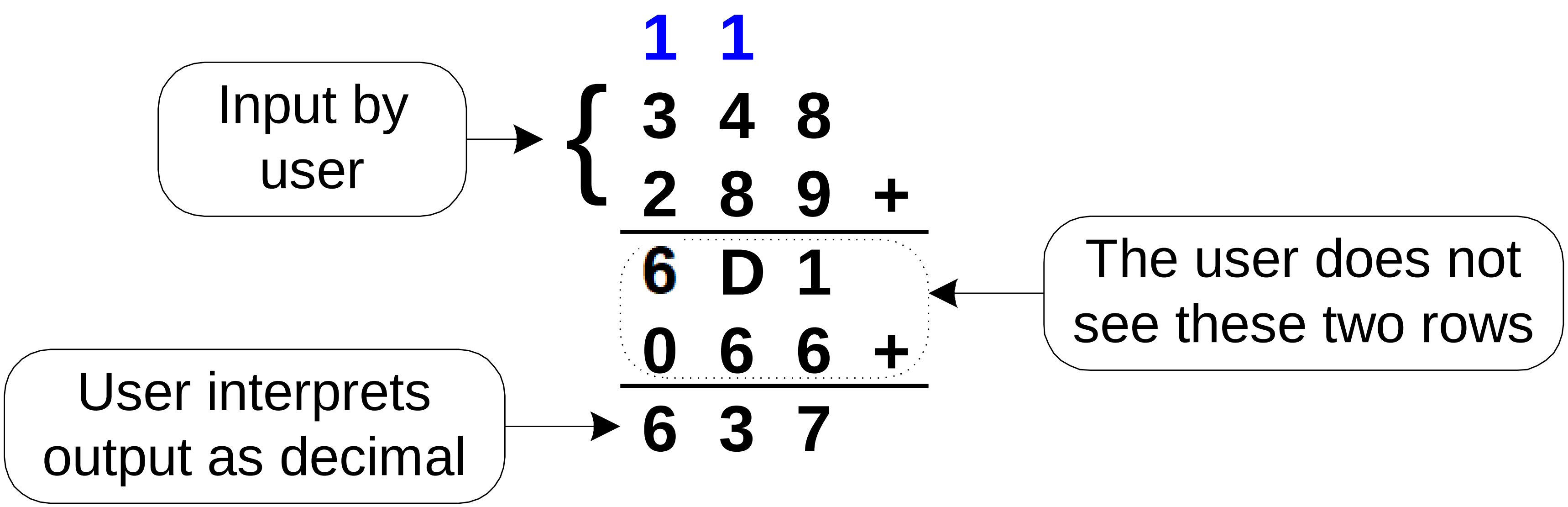
Addition of BCD numbers

Example 1



Addition of BCD numbers

Example 2



Addition of BCD numbers

Some further examples:

	1		
4	5	8	
5	2	6	+
9	8	E	
0	0	6	+
9	8	4	

4	7	5	
5	0	3	+
9	7	8	
0	0	0	+
9	7	8	

1	1		
3	6	8	
5	3	7	+
9	A	F	
0	6	6	+
9	0	5	

1	1	1		
	7	4	7	
	5	6	9	+
1	D	B	0	
0	6	6	6	+
1	3	1	6	

1	1			
	2	5	3	
	7	7	6	+
1	A	C	9	
0	6	6	0	+
1	0	2	9	

1	1			
	9	8	2	
	6	4	7	+
1	0	C	9	
0	6	6	0	+
1	6	2	9	

Addition of BCD numbers

More examples!

1	1	1	1		
	7	8	7	3	
	8	5	8	7	+
1	0	E	0	A	
0	6	6	6	6	+
1	6	4	6	0	

1			1		
	3	1	0	9	
	7	0	8	7	+
1	A	1	9	0	
0	6	0	0	6	+
1	0	1	9	6	

1	1	1	1		
2	5	7	9	6	
	7	6	0	7	+
3	D	E	A	D	
0	6	6	6	6	+
3	3	4	0	3	

The End